

NABILA NEWAZ

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PROFESSIONAL SUMMARY

Computer Science & Engineering undergraduate with hands-on experience spanning applied AI/ML research and full-stack software engineering. Published research on activation steering for LLM reasoning, built RAG pipelines and LLM-driven applications (Gemini, Llama, LangChain, FAISS), and shipped production web and mobile applications across React/Next.js, Node.js/FastAPI, and PostgreSQL/MongoDB. Comfortable across the ML lifecycle and the full engineering stack, with a structured, detail-oriented approach to validating both model outputs and shipped features.

EDUCATION

Bachelor of Science in Computer Science and Engineering

Islamic University of Technology (IUT), Gazipur

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Mining, Digital Image Processing, Pattern Recognition, Data Structures, Algorithms, Database Management Systems, Probability and Statistics, Linear Algebra, Operating Systems, Computer Networks, Theory of Computing, Algorithm Engineering

Higher Secondary Certificate (HSC)

Rajuk Uttara Model College, Dhaka

Secondary School Certificate (SSC)

Rajuk Uttara Model College, Dhaka

RESEARCH EXPERIENCE

Undergraduate Thesis: Latent-Space Steering in Compressed & Continuous Chain-of-Thought Reasoning

Islamic University of Technology (IUT)

- Built and ran a four-phase, test-set-isolated research pipeline across three Qwen2.5 backbones and two reasoning architectures (LoRA-based compressed-text CoT and a COCONUT-style continuous latent CoT), measured by zero data leakage verified across all isolated train, steering, validation, and locked test splits.
- Recovered accuracy lost to reasoning compression on the official 1,319-example GSM8K test set, measured by a 69.0% to 72.0% accuracy gain on Qwen2.5-Math-1.5B and consistent gains reproduced across all three backbones and both architectures, by extracting and applying multi-layer difference-of-means activation-steering directions from labeled hidden states.
- Diagnosed a null result from an initial single-layer protocol and redesigned the extraction/injection methodology, measured by the revised protocol correcting 15.2% of baseline-wrong examples with no increase in reasoning token budget, validated against random-noise, shuffled-label, and head-level Inference-Time-Intervention controls, and shown to transfer from GSM8K to SVAMP without retuning.

Stack: Python · PyTorch · HuggingFace · LoRA · Qwen2.5 · LLMingua-2 · GSM8K · SVAMP

EXPERIENCE

AI / ML Intern

EATL

- Delivered a stable, production-ready AI study planning system for real student users, measured by successful release with no critical post-launch defects, by designing and implementing the system's core planning logic end-to-end.
- Strengthened stakeholder confidence in system outputs, measured by clean sign-off with no unresolved critical issues, by translating technical implementation and validation results into clear, non-technical status reports.

Mobile App Development Intern

Excelerate (Virtual, International Team)

- Delivered Flutter features on schedule across structured sprint cycles, measured by consistent on-time sprint completion, by developing, self-testing, and debugging cross-platform functionality prior to handoff for integration.
- Reduced integration-stage defects, measured by fewer issues surfacing after merge, by identifying and resolving bugs across time-zone-distributed development and review cycles.

Data Visualization Associate Intern

Excelerate (Virtual, International Team)

- Improved data-driven decision accuracy for stakeholders, measured by dashboard outputs consistently matching source data with no unresolved discrepancies, by building and validating dashboards against structured datasets.

KEY PROJECTS

PaperVault: AI-Powered Research Paper Management Platform

- Designed prompt chains for automated summarization, flashcard generation, and semantic paper recommendations, measured by consistently relevant outputs across varied papers, by iterating on prompt structure and retrieval context fed to a Groq-hosted LLM.
- Delivered a live, LLM-integrated research platform, measured by successful automated builds on every push to main, by deploying the summarization and recommendation pipeline behind a CI/CD-backed backend (Vercel, Supabase).
- Implemented secure user authentication and access control, measured by access-control behavior matching intended rules across user scenarios, by configuring Supabase Auth (JWT-based) alongside Row-Level Security (RLS) policies on the Postgres database.

Stack: Groq API · React · TypeScript · Vite · Supabase (Postgres, Auth, Storage, Edge Functions) · Vercel CI/CD

Personalized Study Planner: AI-Powered Application

- Built a Retrieval-Augmented Generation (RAG) pipeline for personalized academic planning, measured by outputs consistently matching expected planning logic, by designing and tuning prompts and workflows across Gemini and Llama (open-source LLM) against defined test scenarios.
- Led end-to-end development of a live, user-facing AI application, measured by successful deployment with no critical defects at launch, by architecting the system across FastAPI, React, and a FAISS-backed retrieval layer.

Stack: FastAPI · Python · React · Tailwind CSS · Gemini · Llama · FAISS

BizAssist: AI-Powered Business Strategy Platform

- Designed a multi-step Gemini-AI evaluation pipeline that scores business ideas across market potential, feasibility, scalability, and profitability, measured by successful end-to-end delivery, by structuring prompts and chaining model outputs across sequential evaluation stages.

Stack: Gemini AI · Next.js · TypeScript · Node.js · Express · Firebase · Cloudinary

Shondhan: AR-Assisted Property Discovery Mobile Application

- Built a cross-platform mobile app with real-time map-based search and AR visualization, measured by consistent accuracy across test devices, by developing and debugging core features using Flutter, Firebase, and ARCore.

Stack: Flutter · Firebase · Google Maps API · ARCore

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C, C++, Java, Dart, HTML, CSS

AI / ML: LLMs (Gemini, Llama, Groq, Qwen), LangChain, Prompt Design, RAG Pipelines, Activation Steering, LoRA, FAISS (Vector Database), PyTorch, TensorFlow, Keras, HuggingFace

Computer Vision: CNNs, OpenCV

Frontend: React, Next.js, Flutter, Tailwind CSS

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST API design & validation, JWT-based authentication

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Firestore, SQL

DevOps & Cloud: Vercel CI/CD, Docker, Supabase (Postgres, Auth, Storage, Edge Functions), Git, GitHub

Tools: VS Code, Postman, Jupyter Notebook, Kaggle, Colab, Figma, Excel, Cloudinary

ACHIEVEMENTS & ACTIVITIES

- Top 12: KUET BitFest Datathon
- 4th Place: CalorieQuest Machine Learning Competition
- Participant: SOLVIO AI Hackathon (Sheba Platform)
- Participant: UIU CSE Fest 2025 Project Show
- Member: IUT Computer Society
- Mentee: SheSTEM Mentorship Program